Title Slide

- **Team Name : NeuroBreath**
- **Problem Title & ID : Real-Time Edge AI Lung Sound Anomaly Detection Using FPGA Acceleration (HW-3)**



Problem Understanding

Your interpretation of the given problem

- Detecting lung sound anomalies (wheezes, crackles, stridor) in real-time at the edge.
- Traditional methods rely on bulky, slow, or cloud-dependent systems.
- Need for low-latency, portable, and reliable detection to assist frontline healthcare.
- High variability in lung sounds: Anomalies differ across age groups, health conditions, and recording environments, demanding adaptive detection.

Key constraints

- Edge-first design: Processing happens locally, no dependency on cloud.
- FPGA acceleration: Ensures high throughput, parallelism, and energy efficiency.
- Noise resilience: Must handle real-world conditions.
- Resource limits: Compact hardware, constrained power, and memory footprint.
- Clinical relevance: Assumes annotated datasets and medically validated sound features.

Why it matters?

- Early diagnosis: Enables faster detection of respiratory disorders like asthma, COPD, pneumonia.
- Accessibility: Portable, low-cost solution for rural and resource-limited settings.
- Scalability: Can integrate into stethoscopes, wearables, or IoT healthcare devices.
- Impact: Bridges AI + hardware for life-saving, real-world healthcare innovation.

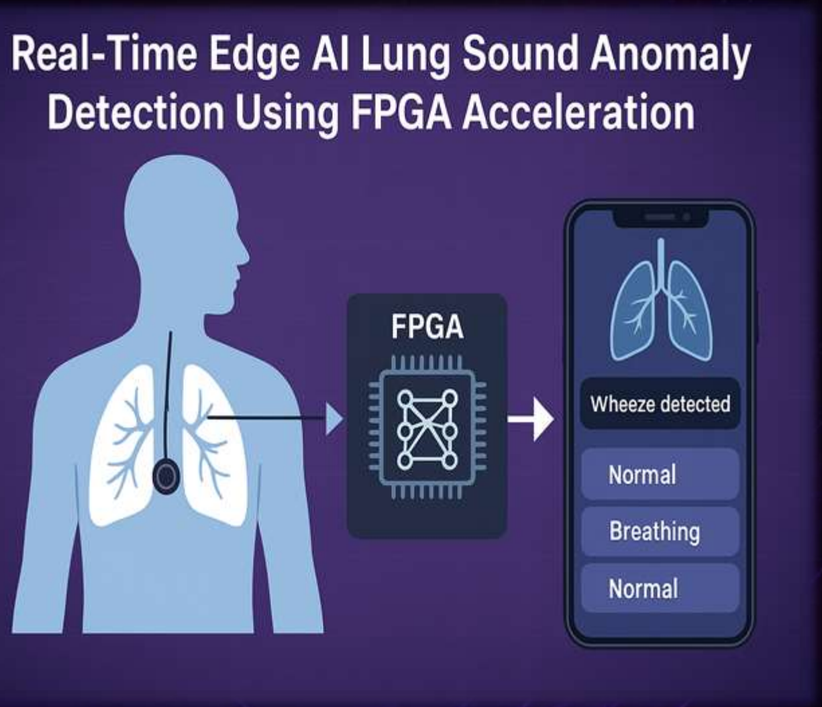
Solution Overview

Your solution proposal

- An FPGA-accelerated edge AI system for real-time lung sound anomaly detection (wheezes, crackles, stridor).
- Compact, portable device that continuously monitors respiratory health with low latency and high reliability.
- Integrated with a mobile app for visualization, alerts, and clinical insights.
- Designed for frontline healthcare: wearable acoustic sensor + FPGA inference engine.

How it addresses the problem at a high level?

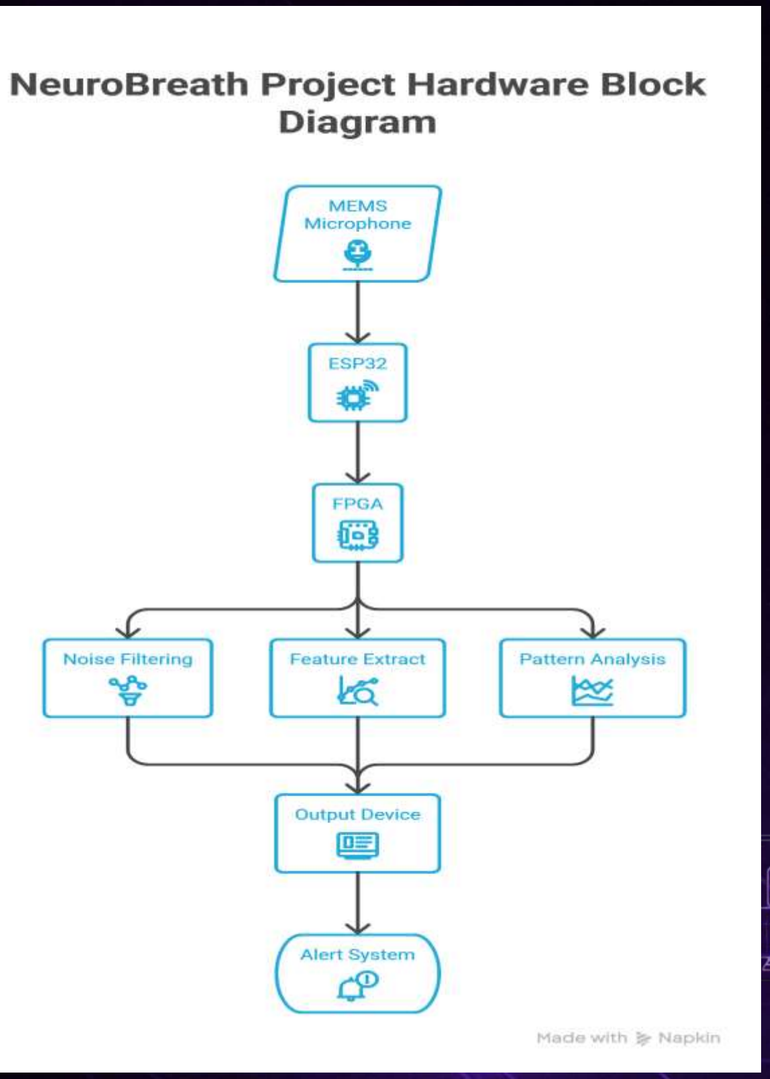
- Enables **early detection** of respiratory disorders by analyzing lung sound patterns in real time.
- Reduces dependence on bulky, cloud-based, or subjective manual assessments.
- Provides **continuous monitoring** with adaptive noise filtering for real-world environments.
- Leverages FPGA acceleration for parallelism, energy efficiency, and ultra-low latency



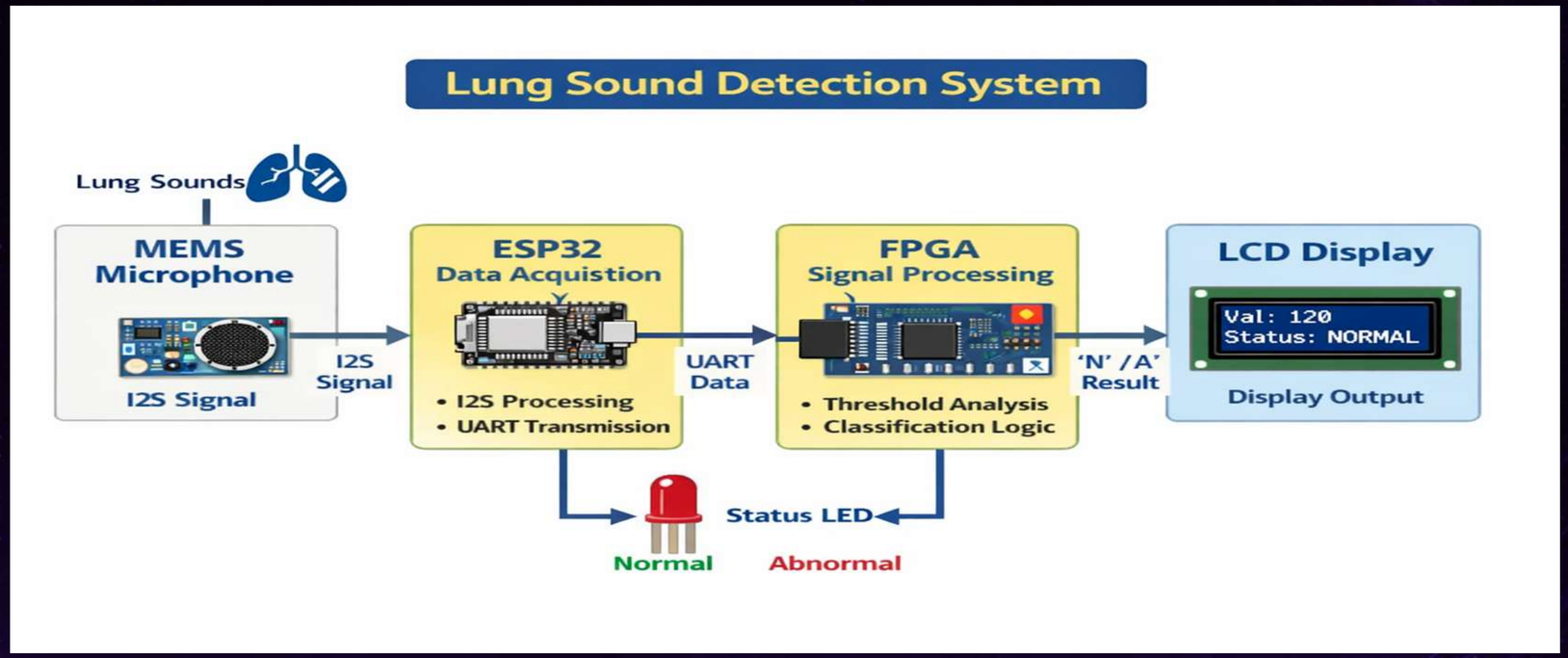
SPECTRE

System Architecture / Block Diagram

Flow chart



Inputs →Processing →Outputs

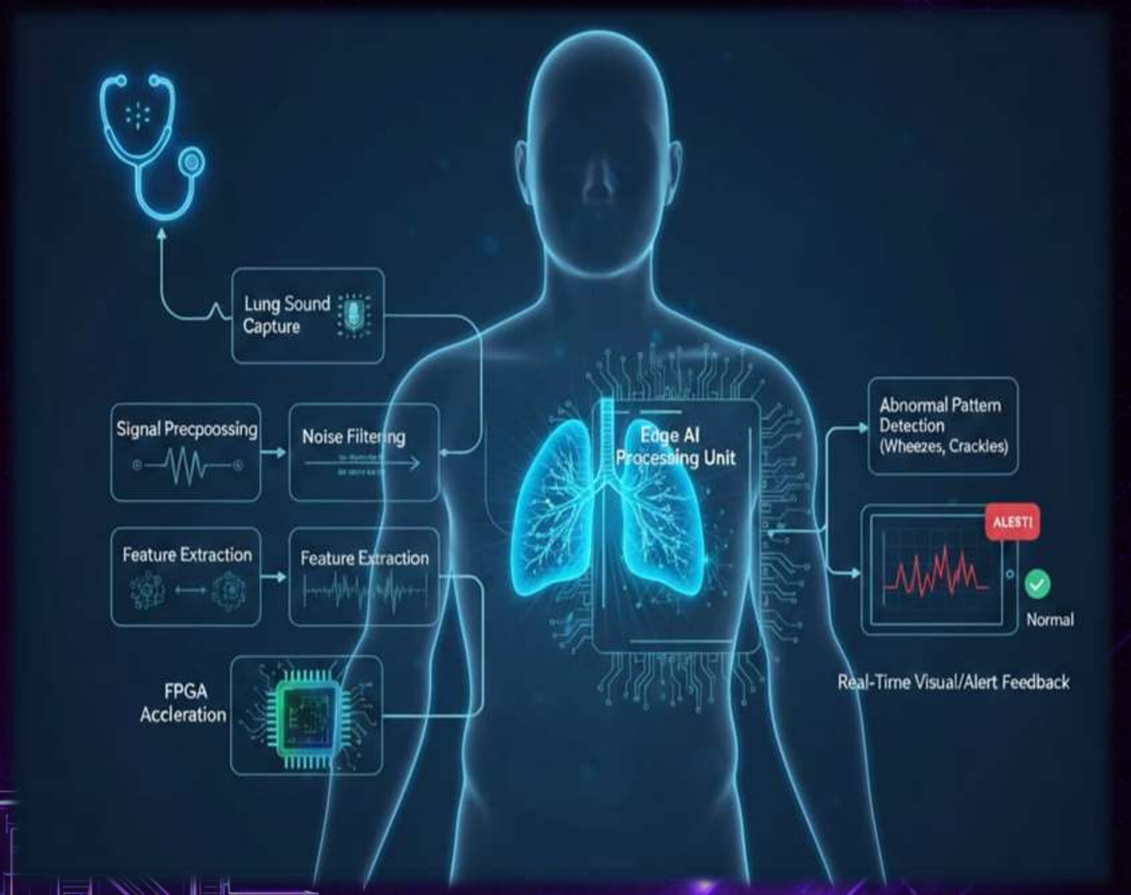


SPECTRE

Methodology / Workflow

Detailed Step-by-step technical approach

- Respiratory Sound Capture: Wearable acoustic sensor continuously records lung sounds (wheezes, crackles, stridor).
- Signal Preprocessing: Noise filtering, segmentation, and normalization to handle real-world variability.
- Edge AI Processing (FPGA): Real-time feature extraction and anomaly detection accelerated by FPGA parallelism.
- Adaptive Classification: AI models trained on annotated lung sound datasets classify anomalies with high accuracy.
- Secure App Integration: Results synced to a mobile app for visualization, alerts, and clinical insights.
- Workflow Fit: Designed to integrate seamlessly into stethoscopes, portable monitors, or IoT healthcare devices



Methodology / Workflow

Logic Flow

- Signal Processing Pipeline
- Acoustic Sensor Input: Captures lung sounds continuously (wheezes, crackles, stridor).
- Band-Pass Filtering: Removes ambient noise and isolates respiratory frequency bands.
- Breathing Cycle Detection: Identifies inhale/exhale phases and irregular patterns.
- Feature Extraction: Computes spectral energy, zero-crossing rate, MFCCs, and breathing rate.
- FPGA-Accelerated Inference: Runs lightweight anomaly detection models in parallel for ultra-low latency.

- Decision Logic
- Anomaly Trigger: If abnormal sound features + irregular breathing rate → raise alert.
- Normal Pattern: If no anomaly → update wellness dashboard with current vitals.
- Contextual Insights: App provides real-time feedback, trends, and preventive suggestions.
- Clinician Mode: Enables deeper analysis and export of annotated sound segments for review.

SPECTRE

Technology & Value Proposition

Tech Stack (hardware + software)

HARDWARE

•Acoustic Sensor (MEMS Mic) – Captures lung sounds continuously.
•FPGA Board (SoC/Accelerator) – Real-time parallel AI inference with low latency.

SOFTWARE

•Signal Processing – Noise filtering, segmentation, and feature extraction (MFCCs, spectral energy).
•Edge AI / ML Models – FPGA-accelerated anomaly detection for wheezes, crackles, stridor, (from datasets)
•Mobile App (Android/iOS) – Real-time alerts, dashboards, and clinician review.
•AI Health Assistant – Provides contextual insights and preventive guidance

What makes this solution better / unique

Real-Time Edge AI: Ultra-low latency detection without cloud dependency.

FPGA Acceleration: High throughput, energy-efficient, and scalable for healthcare IoT.

Continuous Respiratory Monitoring: Detects anomalies early, even in noisy environments.

Clinical Workflow Fit: Seamlessly integrates into stethoscopes, wearables, or portable devices

Privacy-First Design: Edge-first processing, encrypted data, minimal cloud exposure.

Impactful Innovation: Bridges AI + hardware for life-saving respiratory diagnostics in resource-limited settings.